

The Black Box Concept

Encapsulation is the practice of bundling data (variables) and methods into a single unit, while restricting direct access to components. It hides the internal state and requires all interaction to go through public "gatekeepers."

Step 1: RESTRICT DATA

Data Hiding

Use the `private` modifier to ensure specific variables cannot be reached from outside the class.

```
public class Person {  
    private String name; // Hidden from other classes  
}
```

Getters & Setters

Public methods provide a safe way to **read** (Getter) and **write** (Setter) to private fields.

```
// GETTER (Read)
public String getName() {
    return name;
}

// SETTER (Write)
public void setName(String newName) {
    this.name = newName; // 'this' points to the class field
}
```

Why Encapsulate?

- **Control:** You can validate data in the setter (e.g., check if age > 18) before saving it.
- **Flexibility:** Changing internal variable names won't break other parts of the program.
- **Security:** Protects fields from being altered by external malicious or accidental code.

Read-Only vs. Write-Only

By omitting the setter, you create a **Read-Only** field. By omitting the getter, you create a **Write-Only** field.